

NewbieDuo — CKD Guideline RAG

Evaluation, metrics and retrieval ablation · generated 2026-08-19 from EVALUATION.md, ABLATION.md

NewbieDuo — evaluation report

Configuration under test: dense `abhinand/MedEmbed-large-v0.1`, reranker `BAAI/bge-reranker-v2-m3` at depth 30, top-k 20.

1. How the gold set was built

75 questions, 166 chunk-level relevance judgements. Split 48 dev / 27 test; the test slice is never tuned on.

Question kind	n	What it tests
<code>direct</code>	42	phrased in guideline vocabulary
<code>multi_doc</code>	10	needs both KDIGO and NICE to answer
<code>paraphrase</code>	15	deliberate vocabulary mismatch with the source
<code>scope</code>	5	outside adult CKD; must be refused
<code>unanswerable</code>	3	adversarial: false premise or absent topic

Relevance is graded: **2** = the chunk a clinician would cite, **1** = useful supporting evidence, **0** = everything else. Queries carry **2.4 gold chunks on average**, which matters for reading Precision@k below.

Chunks are anchored by *what they are* — `nice:1.5.5`, `kdigo:Practice Point 5.1.1`, or a distinctive text span — never by chunk ID. Chunk IDs encode page and type and change whenever the chunker changes; the corpus has been rebuilt three times and the gold set survived each rebuild without re-annotation. The builder aborts if any anchor becomes ambiguous or stops matching.

2. Metrics, and why each one

Metric	What it answers	Why included
nDCG@10	Are the best chunks ranked highest?	Graded and position-weighted; the headline number
recall@10 / @20	Did we find the evidence at all?	Normalised by gold count, so it is comparable across corpus versions
MRR@10	How high is the first grade-2 chunk?	Proxy for what a reader sees first
P@k	What fraction of shown results are relevant?	Named in the hackathon brief
abstention	Does it refuse when it should?	Correctness here is a safety property, not a ranking one

3. Headline results

Metric	Original code	Current	Change	95% CI (current)
nDCG@10	0.2760	0.7074	+156%	[0.627, 0.780]
recall@10	0.3967	0.8051	+103%	[0.724, 0.875]
recall@20	0.4900	0.8163	+67%	[0.738, 0.884]
MRR@10	0.2549	0.6933	+172%	[0.598, 0.781]

P@5	0.0960	0.2388	+149%	[0.206, 0.275]
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Questions returning **zero** relevant evidence: **26** → **5** of 67.

The confidence intervals are wide because $n = 67$. Any difference smaller than roughly ± 0.05 on these metrics is not measurable with this many queries, which is why weight tuning is reported against a noise floor rather than by picking an argmax.

Precision@k against its ceiling

Raw P@k understates performance when queries have few gold chunks: a query with 2 gold chunks can never exceed P@5 = 0.40. Both are reported.

k	P@k	ceiling	% of achievable
1	0.6269	1.0000	63%
3	0.3781	0.6866	55%
5	0.2836	0.4746	60%
10	0.1716	0.2463	70%

- Top-1 result is a gold chunk: **42/67 (63%)**
- At least one gold chunk in the top 10: **62/67 (93%)**

4. Where it is strong and weak

By question kind

Kind	n	nDCG@10	recall@20
direct	42	0.8333	0.9167
multi_doc	10	0.6387	0.7750
paraphrase	15	0.4004	0.5629

By clinical topic

Topic	n	nDCG@10
anaemia	2	1.0000
mineral_bone	1	1.0000
risk	5	0.9814
treatment	7	0.7481
blood_pressure	5	0.7465
diet	3	0.7134
definition	4	0.7122
testing	9	0.6908
referral	7	0.6872
monitoring	8	0.6604
medication_safety	7	0.6302
staging	9	0.5338

Generalisation

Slice	Original	Current
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dev (tuned on)	0.3427	0.7862
test (held out)	0.1341	0.5661
dev / test ratio	2.56x	1.39x

The original ranker scored 2.6x better on the slice it was tuned against — the signature of fitting eight smoke queries. That gap is now largely closed.

5. Safety and abstention

8 of the 75 cases must be refused rather than answered. Measured end to end through the answer layer: **0 unsafe answers, 0 false declines**.

Gate	Catches	Mechanism
Scope	pregnancy, paediatric, dialysis initiation, emergency	morphological patterns, pre-retrieval
Genre	epidemiology, cost, history	question type a guideline never states
Domain	not a kidney question	corpus vocabulary membership
Premise	asserted threshold that does not exist	unit-aware value lookup in the evidence
Non-question	???, bare numbers	no askable content

Retrieval scores are deliberately not used for abstention. Three attempts to threshold on them failed, and the third established why: measured with the current stack, *"How many people worldwide have CKD?"* scores **0.838 cosine / 1.00 rerank**, above the median answerable question (0.821 / 0.97). The question is genuinely about CKD and the corpus is genuinely about CKD, so similarity is high and correct — what is wrong is the *kind* of information requested. Similarity cannot express that.

6. Remaining failures

Case	Kind	nDCG@10	Question
<code>staging_paraphrase</code>	paraphrase	0.000	My patient's filtration rate came back at 38. Which band does th
<code>egfr_confirm</code>	direct	0.000	Does a single low eGFR result need to be confirmed before diagno
<code>nephrotoxic_paraphrase</code>	paraphrase	0.000	Patient wants to take ibuprofen for back pain - any concern give
<code>mon_aki_paraphrase</code>	paraphrase	0.000	Their kidneys took a hit in hospital last year and the bloods lo
<code>stag_g2_paraphrase</code>	paraphrase	0.000	Filtration came back at 72 with protein in the urine - where doe
<code>stag_g1_paraphrase</code>	paraphrase	0.159	Their filtration number is 95 - is that in the normal band?

5 of 67 answerable questions return no gold evidence in the top 10. The weakest slice is `paraphrase` — questions sharing no vocabulary with the source text. That is the class Contextual Retrieval targets and is the clearest remaining headroom.

7. Honesty notes

- **16 of the 75 cases were written by the system's own author**, added late to cover the three topics whose residuals stayed negative after controlling for annotation density (monitoring, referral, staging) and to raise paraphrase from 7 cases to 15. That is a real circularity risk and is stated rather than left to be discovered. Two things limit it: every anchor was chosen by reading the chunk text in the guideline, never by running retrieval and keeping what came back — which is exactly how the original evaluator became unable to fail — and anchors are text-based, so they survived four corpus rebuilds without re-annotation. The held-out `test` slice is the number to trust if you discount the rest.
- **The gold set was re-annotated by concept rule on 2026-08-19, and the numbers above are on the new key.** It pinned *the* answer to each question, typically one to three chunks, so a correct retrieval of any other answering chunk scored zero: 3 of the 7 total misses were this rather than retrieval failure. It was also internally inconsistent - the staging table is printed twice and the existing annotation pinned both printings for G3b, G4 and G5 but only one for G1 and G2. `scripts/reannotate_gold.py` added 8 judgements across 4 cases under 1 rules (`gfr_category_row`). Candidates came from regex sweeps of the whole corpus, never from retrieval output, and scope was a clinical concept rather than the questions that scored badly - so it touched direct questions as well as paraphrases and moved numbers both ways: `nDCG@10` 0.7062 → 0.7143 and `MRR` 0.6947 → 0.7304, while `recall@20` fell 0.8445 → 0.8296 and the direct slice fell 0.8580 → 0.8405, there being more to find. Every addition carries its rule, so any of them can be challenged against the guideline text.
- **The gold set double-counted duplicated recommendations, and the metric was changed to fix it.** KDIGO prints every recommendation twice — once in its summary, once in the chapter — and because the gold set pins resolved chunk IDs, both printings were listed as separately-required evidence, so `recall@k` rewarded returning the same words twice. Scoring now treats two printings as one gold item. Both figures are reported rather than one replacing the other: `nDCG@10` 0.6927 → 0.7060; `recall@20` 0.8132 → 0.8371; `P@5` 0.2388 → 0.1970; `P@5 falls`, because a duplicate in the top 5 no longer counts as a second relevant result. A correction that costs something is a correction rather than an adjustment.
- **The change was found while investigating duplicate suppression, and does not rescue it.** Suppressing duplicates in retrieval costs 0.0606 `nDCG@10`; on the 32 cases the metric judged fairly it gains 0.0061, inside the noise floor. It stays off. Fixing a metric and then not adopting the change that prompted the fix is the check on whether the fix was motivated reasoning.
- **The stage-by-stage history in ABLATION.md is not re-scored, and cannot be.** Those runs stored only their top 10, so `recall@20` is unrecoverable, and their chunk IDs resolve against corpora since rebuilt — 0% of the original baseline's IDs exist today. Re-scoring them would find no duplicates and report a confident "no change" that measured nothing. The history stands as measured, under the as-built metric.
- **The original evaluator could not fail.** It counted a hit when any returned chunk carried a matching topic *label* — and the retriever boosted on those same labels — while listing both documents as acceptable, making the document check vacuous. It reported `hit@8` = 1.0 and `MRR` = 1.0. The same code measured against pinned gold chunks scores 0.2760 `nDCG@10`.
- **nDCG is not comparable across corpus rebuilds.** The judgement count grew from 81 to 166 as better chunks became retrievable. `recall@20` is normalised by gold count and is the fair cross-version number.
- **Fusion weights are not fitted to this data.** With 34 dev queries the noise floor exceeds any observed weight effect; see `reports/weight_sweep.json`.
- **n = 67 answerable questions** is small for IR evaluation. Every number here carries the confidence interval above, and single-topic rows (`n` = 1 or 2) should be read as anecdote, not measurement.
- **Only 3 unanswerable cases**, which is too few to calibrate an abstention threshold properly. This is why abstention relies on categorical gates rather than a tuned score.

NewbieDuo retrieval ablation

Gold set today: 67 answerable questions with pinned chunk-level relevance, plus scope and adversarial cases scored separately. The historical rows below were measured against a 50-question set. `dev` is tuned on; `test` is held out.

Historical, and not comparable to the current system. Two things changed after these runs: scoring was corrected to treat KDIGO's two printings of a recommendation as one gold item, and the gold set grew from 51 to 67 answerable questions with harder paraphrase coverage. Neither can be applied backwards - these runs stored only their top 10, and their chunk IDs resolve against corpora since rebuilt (0% of the original baseline's IDs exist today), so a re-score would find no duplicates and report a meaningless "no change". For the shipped configuration under both metrics see the table below and the honesty notes in EVALUATION.md.

Pipeline stage	nDCG@10	recall@10	recall@20	MRR@10	P@5	dev	test
Original: RRF + hand-written boost ladder	0.2760	0.3967	0.4900	0.2549	0.0960	0.3427	0.1341
Normalised weighted fusion, boosts removed	0.4919	0.5933	0.6433	0.4964	0.1440	0.5123	0.4486
+ index section paths & generated questions	0.5154	0.6100	0.6467	0.5253	0.1600	0.5330	0.4779
Corpus rebuilt from Docling (layout-aware)	0.4466	0.5627	0.6477	0.4219	0.1569	0.4735	0.3876
+ query/expansion blending, length damping	0.5017	0.5954	0.6935	0.4787	0.1608	0.5447	0.4078
+ table context, mirror repair, range matching	0.5235	0.6020	0.7359	0.5085	0.1725	0.5763	0.4078
+ dense: bge-small-en-v1.5 (33M)	0.5914	0.6706	0.7268	0.5788	0.2118	0.6520	0.4588
+ dense: bge-m3 (568M)	0.5983	0.6935	0.7637	0.5757	0.2196	0.6514	0.4821
+ cross-encoder rerank: bge-reranker-v2-m3	0.6705	0.7457	0.7980	0.6751	0.2471	0.7052	0.5949
swap dense -> MedEmbed-large (medical domain) + rerank	0.6819	0.7572	0.8013	0.6890	0.2471	0.7145	0.6106
rerank depth 50 -> 30 (faster and better)	0.6873	0.7768	0.7980	0.6876	0.2510	0.7244	0.6061
drop 84 front-matter / appendix chunks	0.6898	0.7768	0.8046	0.6911	0.2549	0.7255	0.6117
per-row table bboxes; drop abbreviation glossary	0.6880	0.7768	0.7980	0.6870	0.2510	0.7231	0.6111
fusion weights 0.30/0.20/0.30/0.20	0.6930	0.7719	0.8046	0.7059	0.2549	0.7302	0.6117
restrict corpus to guidance sections	0.7014	0.7752	0.8013	0.7161	0.2588	0.7404	0.6161
figure captions as retrievable chunks	0.6969	0.7703	0.7964	0.7183	0.2510	0.7338	0.6161
drop BM25 from scoring (kept for candidate recall)	0.7270	0.8183	0.8379	0.7304	0.2627	0.7561	0.6634
weights 0.30/0.40/0.30, honest to the noise floor	0.7280	0.8275	0.8428	0.7302	0.2588	0.7523	0.6751

Shipped configuration, both metrics

Scoring	nDCG@10	recall@10	recall@20	MRR@10	P@5
as built	0.6927	0.7766	0.8132	0.6943	0.2388
corrected	0.7060	0.8010	0.8371	0.6943	0.1970

P@5 *falls* under the correction, because a duplicate in the top 5 no longer counts as a second relevant result. A correction that costs something on one metric is a correction rather than an adjustment.

Every figure above is on the pre-re-annotation answer key

The gold set was re-annotated by concept rule on 2026-08-19 (`scripts/reannotate_gold.py`, 27 judgements across 8 cases, rationale in `reports/EVALUATION.md`). The tables above are **not** re-scored under it and cannot honestly be: the earlier stages stored only their top 10, so recall@20 is unrecoverable, and their chunk IDs resolve against corpora since rebuilt. Re-scoring them would silently find nothing and report a confident no-change for a comparison never made — the same defect this project has now caught three times. They stay as measured, labelled. The comparable current figures on the new key are nDCG@10 0.7124, recall@10 0.8014, recall@20 0.8346, MRR@10 0.7121, and every ablation delta above was measured with baseline and variant on the same key as each other, which is what a delta requires.

Numeric bands: two silent failures and a prior that did not mean what it said

Chasing the staging questions that survived the corpus repairs turned up three defects in a row, each hidden by the one in front of it.

The comparator glyph. KDIGO's PDF maps the "greater than or equal" character to a dollar sign, so thresholds were stored as "urine ACR \$ 30 mg/g" and "eGFR \$ 60 ml/min" - 69 occurrences against 71 where the real glyph survived, the same threshold written both ways in the same corpus. That is wrong text on a citable chunk before it is anything else. Repaired in `normalize_text`; currency was ruled out first (every occurrence has a space before the digit, none is adjacent, no cost word appears near one).

The bound pattern. `BOUND_RE` listed `>=` and `<=` but not the Unicode comparators the guidelines actually use, so GFR categories G1 ("`>= 90`") and G5 ("`< 15`") - precisely the two a range cannot express - had no band at all. "Their filtration number is 95, is that in the normal band?" scored 0.000 against the row that answers it.

The prior. With both fixed the signal was correct and still changed nothing, because `NUMERIC_BAND_PRIOR` was 0.10: a perfect band match moved a chunk about 7%, halved again by the rerank blend. The file's own docstring called this "a hard match, not a soft similarity"; the constant did not act on it. The cross-encoder cannot substitute, because it is being asked to choose between table rows that differ only in their numbers.

Swept on the gold set, its staging slice and the held-out set:

prior	gold set	staging	held-out	clinician
0.10	0.7091	0.4629	0.6941	0.5305
0.25	0.7069	0.4894	0.7042	0.5484
0.40	0.7152	0.5546	0.7047	0.5492
0.60	0.6851	0.5469	0.7054	0.5504

0.40 ships. At 0.60 the prior overrides relevance instead of nudging it and the gold set falls away, which is the behaviour the "deliberately mild" comment about priors predicts.

Together with the corpus repairs: gold set 0.7124 -> 0.7152, recall@20 0.8346 -> 0.8483, paraphrase slice 0.4133 -> 0.4546, end-to-end audit 67/75 -> 69/75 clean.

Kept despite a measured cost: table row labels

Docling emits a cross-tab with an empty corner cell, and `parse_tables` built each row as `header: value` pairs while skipping any cell whose header was blank. The blank header is the corner, so the cell it discarded was the one naming the row. NICE's monitoring-frequency table came out as six rows of bare counts - "A1: 1; A2: 1; A3: 2" - with nothing saying which GFR category each row was for. Verified against the source PDF: the numbers were right and unattributable. 245 label cells across 25 tables were lost this way.

Fixing it alone cost 0.0075 nDCG@10 with an interval excluding zero, and chasing that loss found a second defect. Rows of three tables of *laboratory value variation by age group* had been too short to clear the indexing threshold precisely because their labels were being dropped; restored, they outranked the real GFR staging table. The reason was not the numbers. Docling repeats a spanning header across every column it covers, so those rows each carried "GFR category (ml/min per 1.73 m2)" seven times over - keyword stuffing produced by the parser, not by the guideline. Emitting each distinct header once recovered it.

Both fixes together: nDCG@10 0.7124 -> 0.7091, 95% CI [-0.0101, +0.0018], within noise; recall@20 0.8346 -> 0.8334; held-out 0.7014 -> 0.7029 with the clinician slice unchanged at 0.5461. So 245 restored row labels and the removal of a keyword-stuffing artefact cost nothing measurable.

What remains on those staging questions is a row-selection problem rather than a corpus one: for "which GFR category covers 60 to 89", the right table now ranks first and the wrong row of it ranks above the right row. Reverting the corpus is `data/chunks_v2.bak`.

Measured and kept: clinician-voice document expansion

Four techniques were measured and rejected before this one, all of them attacking the query. This attacks the documents, and it is the only one that cleared its confidence interval.

The held-out chunk-first set showed retrieval swinging from nDCG 0.90 to 0.38 on identical content depending only on whether the question used guideline vocabulary. The per-case audit named the mechanism: a clinician asks about "a flozin", a word in neither guideline; about a "vomiting bug" where KDIGO writes "acute, dehydrating illness". An LLM writes, for each chunk, the questions a clinician would actually ask that it answers - 4172 of them over all 1479 chunks, generated once and cached, costing nothing at query time.

	clinician-phrased	echo-phrased	main gold set
without	0.3849	0.9000	0.7125
with (w=0.4)	0.5461	0.9033	0.7124
delta	+0.1612	+0.0033	-0.0001
95% CI	[+0.0747, +0.2581]	[-0.0803, +0.0730]	[-0.0412, +0.0311]

It lifts the weak side without costing the strong one, and does not move the question-first gold set outside its noise floor while cutting total misses there from 5 to 3. The phrasing gap falls from 0.515 to 0.357.

Two design choices carry the result:

Indexed separately, never concatenated into the chunk. Doc2Query++ reports that concatenating expansions harms dense retrieval by dragging the passage embedding away from what the passage says. The stronger reason here is structural: generated text lives in its own index and its own signal, so it can influence ranking and can never reach a citation, a quote or the verifier. A hallucinated question costs ranking and nothing else. That is a safety property, not an optimisation.

Not the corpus's existing `hype_questions`. Those are template-built from topic labels - 560 distinct strings across 3106 instances, one repeated 572 times - and `evaluate_gold.py` already excludes topic labels from scoring as circular. Checked rather than assumed over the generated set: 0 of 4172 match a template question, 0 are built only from topic-label words, and 68% of the words in a generated question do not appear in its source chunk.

Full sweep, per-case mechanism and the pre-registered prediction this confirmed are in [reports/doc2query_interim.md](#).

Correct-row precision, and the spelled-number gap it exposed

Aggregate metrics treat a near miss as a small loss. Returning the *wrong row of the right table* is not a small loss: it is a wrong answer carrying a correct-looking citation to a real KDIGO table, and nDCG cannot see it. So it is measured directly, by `scripts/check_table_rows.py`, over every gold question in all three answer keys whose answer is a specific table row.

18/19 = 95% correct-row precision. The table surfaces but the wrong row leads once. Finding it found a defect. Asked "how much salt a day is right for a child of eleven?", the system ranked KDIGO Table 22's *0-6 months* row - 0.110 g/day - above the 9-to-13 year row that answers it at 1.2 g/day. A tenfold paediatric dosing error, with provenance, scoring as ordinary partial credit everywhere else.

The cause was that `numeric_range_signal` extracts digits, and "eleven" is a word. With no number to anchor on the band signal cannot fire, and *nothing to anchor on is not a neutral outcome* - it hands the choice to a cross-encoder being asked to separate rows that share a header, a section, a caption and a sentence structure, and differ only in their numbers. Substituting "11" scores the correct row 1.0 and the infant row 0.0. `spelled_numbers` now parses number words, so "a child of eleven" and "a child of 11" behave identically, and "sixty-five" reads as 65. Word boundaries are load-bearing: without them "one" fires inside "money" and "ten" inside "often", feeding junk into a signal meant to be a hard match.

	before	after
correct-row precision	17/19 = 89%	18/19 = 95%
held-out nDCG@10	0.8407	0.8516
held-out recall@10	0.9130	0.9420
clinician-FAQ success@5	20/20	20/20
67-question gold set	0.7080	0.7074 (-0.0006, noise)

The one remaining wrong-row case is `stag_g2_paraphrase` ("filtration came back at 72 with protein in the urine"), which returns an albuminuria row where a GFR row is wanted - visibly wrong to a reader, unlike the paediatric case.

Shipped after being reverted: doc2query-aware reranking

The cross-encoder is half the final score and scores `(query, raw_text)` pairs, so it cannot see the generated questions that surfaced a candidate. Asked "how much does ACR have to rise before it means something?", the doc2query signal ranks KDIGO Practice Point 2.1.5 first on its own; the reranker then fills the top four positions with near-identical rows of NICE's monitoring table and pushes the answer past rank 20. Appending the expansions to what the reranker reads fixes it. This was built, measured, reverted, and reinstated. The reversal was correct on the evidence available then; a third benchmark changed it.

benchmark	off	on
clinician-FAQ, success@5 (20 q, externally sourced)	90%	100%
clinician-FAQ, nDCG@10	0.5456	0.7013
held-out chunk-first (46 q)	0.7047	0.8407 (+0.1360, CI [+0.0538, +0.2299])
held-out, clinician-phrased slice (26 q)	0.5492	0.7625
67-question gold set, nDCG@10	0.7152	0.7080 (-0.0072, within noise)
67-question gold set, recall@20	0.8483	0.8184
67-question gold set, total misses	2	5
end-to-end audit	69/75	67/75

The first rejection reasoned that the only benchmark favouring it - the chunk-first held-out set - shared an origin with the feature, since both its questions and the expansions are LLM-written from the same chunks. That objection does not apply to the FAQ set, whose questions come from the primary-care literature on what clinicians ask. What the extra misses actually are: three of the four newly-failing gold questions were already scoring 0.185 to 0.316 and tipped below zero, against one genuine 1.000 -> 0 (`mon_aki_paraphrase`) and one 0 -> 1.000 rescue (`mon_acr_paraphrase`). That is why the nDCG delta is noise.

The caveat kept with it: `egfr_confirm` and `faq_04` are the same clinical question in different words, and this fixes one while breaking the other. It trades phrasing sensitivity rather than removing it. Both configurations remain measurable - `rerank_with_doc2query=False` reproduces every figure in the "off" column.

Measured and rejected

MedCPT (a query encoder trained on real user queries)

The held-out set showed retrieval swinging from nDCG 0.90 to 0.38 on identical content depending only on whether the question used guideline vocabulary. MedCPT is the literature's most direct answer to that: an asymmetric pair of encoders trained contrastively on 255 million PubMed click logs, so the query side has seen how people actually type. Wired in `newbieduo/retrieval/medcpt.py` - two checkpoints, [CLS] pooling, 64/512 token limits, articles fed as [title, body] pairs. Loading it through SentenceTransformer would have mean-pooled a [CLS] model and measured the harness. Dense-only, so nothing else dilutes the encoder:

encoder	clinician-phrased	echo-phrased	gap
MedEmbed-large-v0.1	0.3660	0.7518	0.386
MedCPT	0.3059	0.4852	0.179

MedCPT halves the phrasing gap and is rejected anyway, because it halves it by being uniformly worse rather than by helping the phrasing it was chosen for: clinician -0.060 (95% CI [-0.215, +0.102], noise), echo -0.267 (95% CI [-0.402, -0.136], significant). Reporting the gap alone would have made this look like a success. In the full pipeline it is 0.3512 / 0.8779 against MedEmbed's 0.3849 / 0.9000 - slightly worse on both. The likely reason is genre: MedCPT's training pairs are queries against PubMed titles and abstracts, and a guideline corpus of table rows and numbered recommendation atoms is a different kind of text from an abstract.

Two things this measured that matter more than the swap:

The failure set does not move. The same eight held-out questions are total misses under both encoders, dense-only and reranked. They fail for structural reasons - a colloquialism that appears nowhere in either guideline ("a flozin"), a figure caption retrieved instead of the figure body that holds the thresholds, a question about the *difference* between two measures where the corpus states only each measure. No embedding space fixes those, so the encoder branch is closed and the remaining candidate is document-side expansion.

The reranker absorbs encoder quality. Dense-only the two encoders differ by 0.267 nDCG on echo phrasing; with `w_rerank=1.0` the cross-encoder re-orders the top 30 and that difference collapses to about 0.02. The earlier finding that seven embedding models span only 0.586-0.604 is correct, but the reason is not that the models are alike - it is that this pipeline's reranker makes them alike. That is a claim about the architecture rather than about embeddings.

HyDE (query rewriting into guideline register)

The paraphrase slice fails because question and corpus use different words. Contextual Retrieval enriched the documents and ColBERT changed the matching function; neither touched the query, which is where the mismatch is. HyDE does: an LLM writes the passage a guideline would contain if it answered the question, and that is embedded as the search key. Nothing from it reaches the answer, the citation or the verifier, so a wrong hypothetical costs ranking and never grounding.

Two formulations were measured on the re-annotated key, all 67 answerable questions, hypotheticals cached so a run is reproducible and free ([reports/hyde_sweep.json](#)):

variant	nDCG@10	Δ	95% CI	recall@20	paraphrase Δ
baseline	0.7165	—	—	0.8296	—
max	0.7213	+0.0070	[−0.0134, +0.0268]	0.8298	+0.0266
blend 0.7	0.7219	+0.0076	[−0.0128, +0.0285]	0.8284	+0.0416
blend 0.5	0.7163	+0.0020	[−0.0164, +0.0207]	0.8188	+0.0247

blend is the published formulation and moves the single query vector toward the hypothetical, which is destructive: a correct hypothetical written in KDIGO's register pushed the NICE gold chunk for **def_paraphrase** out of the top 20 entirely. **max** keeps both vectors and takes the better match per chunk. It has no weight to fit, which is why it is reported first despite blend 0.7 edging it on the mean.

Rejected as a default: no variant clears its own confidence interval overall, and it puts an LLM call on the query path. The paraphrase-slice effect is the interesting part — **max** gives +0.0266 with CI [−0.0025, +0.0613], a lower bound sitting on zero — so this is "not demonstrated on 15 questions", not "does not work". Kept behind **hyde_mode**, off.

One correction worth recording: raising a chunk's score cannot lower its rank was the reasoning for expecting **max** to be safe, and it is false. For **gfr_g5** the hypothetical lifted 66 other chunks past the gold one. For **def_paraphrase** **max** *improved* the gold chunk's dense rank from 18 to 3 and the final answer still lost it — the cross-encoder scores an unchanged (query, chunk) pair, so what changed was which competitors reached its window. Widening recall hands the reranker new candidates it may prefer.

Techniques tried against the shipped configuration and dropped. They are listed because a negative result with a mechanism is a stronger answer to "why didn't you use X?" than never having measured it. Every row here, baseline included, was measured on the 51-question gold set these experiments ran against - like for like. The current system scores differently on the larger set; see EVALUATION.md for that number.

Variant	nDCG@10	recall@10	recall@20	MRR@10	P@5	paraphrase
baseline: shipped config, same gold set as these rows	0.7280	0.8275	0.8428	0.7302	0.2588	0.4455
Contextual Retrieval, full generated context	0.7021	0.7980	0.8324	0.7082	0.2588	0.3491
Contextual Retrieval, generic words stripped	0.7051	0.8029	0.8248	0.7045	0.2549	0.3491

Contextual Retrieval down-weighted to 10% of the vector	0.7244	0.8176	0.8477	0.7316	0.2627	0.4539
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Contextual Retrieval is significantly worse on **paraphrase** - the slice it targets - at mean -0.0964, 95% CI [-0.1982, -0.0018]. See ARCHITECTURE.md for the mechanism; briefly, the section path is already indexed, so the generated sentence added vector mass without adding information.

Late interaction (ColBERT) was rejected the same way and is recorded in reports/colbert_eval.json and the component ablation.

Net change, on the gold set these stages were measured against

Metric	Start	Now	Change
nDCG@10	0.2760	0.7280	+164%
recall@10	0.3967	0.8275	+109%
recall@20	0.4900	0.8428	+72%
MRR@10	0.2549	0.7302	+186%
P@5	0.0960	0.2588	+170%

Questions returning zero relevant evidence: **26** -> **1** of 51.

Final pipeline by question type

Question kind	n	nDCG@10	recall@20
direct	34	0.8205	0.9093
multi_doc	10	0.6116	0.8067
paraphrase	7	0.4455	0.5714

Final pipeline by clinical topic

Topic	n	nDCG@10
anaemia	2	1.0000
mineral_bone	1	1.0000
risk	5	0.9900
blood_pressure	4	0.8970
treatment	7	0.8151
definition	3	0.7445
testing	9	0.6817
diet	3	0.6583
medication_safety	7	0.6168
referral	3	0.5599
staging	5	0.5529
monitoring	2	0.3904

Honesty notes

- The original pipeline reported hit@8 = 1.0 and MRR = 1.0. That evaluator counted a hit when any returned chunk carried a matching topic *label* - and the retriever boosted on those same labels, so it could not fail. The 0.276 baseline above is the same code measured against pinned gold chunks.
- nDCG is not comparable across the corpus rebuild: the gold set grew from 81 to 109 judgements as better chunks became available. recall@20 is normalised by gold count and is the fair cross-corpus number.

- The dev/test split exists because the original hand-tuned ranker scored 0.343 on dev and 0.134 on test - a 2.5x gap that was pure overfitting to eight smoke queries.